tools, such as Chef and Puppet. This doesn't become important until you're scaling, though, and scaling means your server is popular, which is a nice problem to have.

Packages in Python are not a solved problem. There are a variety of tool kits out there that do this and help deal with dependency management. As your project changes over time, you need to ensure your production environment continues to meet the dependency requirements of your project. Failure to do so might cause bugs in production. In some cases, it could cause the project to crash, resulting in a `500` error. Tools like Pipenv, Poetry, and Hatch help manage package dependencies, and each has its own strengths and weaknesses.

B.8.2 Monitoring

Even when everything goes well and your code gets happily deployed, there is a chance of a failure some time in the future. Servers run out of disk space, shared VPS environments have their CPU monopolized by bad neighbors, and cosmic particles rain through the ether and blow up processors and disk drives. That last one doesn't happen very frequently, but it has been known to happen.

Once your server is running, you want to make sure it stays running. Or, at least, you want to be notified if it stops running. Monitoring is the process of keeping an eye on your running systems. A simple version of this might be a shell script that uses `curl` once a minute to hit your web server, and if it gets no response, it kills and restarts your process.

Web services like Uptime and Pingdom monitor your pages and notify you when a page becomes unreachable. Most monitoring sites even hit your project from a variety of places on the planet and log information about the response time. If you end up with a lot of users in a foreign country, you might consider deploying a local server for them to reduce their response time. These kinds of tools only provide an outside view, a binary Service Is Up or Service Is Down perspective, but they tend to be inexpensive. Your hosting provider may offer more detailed services that are integrated with notification tools.

PagerDuty, Dynatrace, and other services allow you to add calls to their API within your code. This is somewhat like logging but with a view to monitoring your system. Alarms can be set up for different events as well as when events, like a server heartbeat, are not received in a timely fashion.